

Professional Agentic Architect

A self-paced study book for Google Cloud PR000340 (beta, 2026-09):
custom agents, evaluation, deployment, security, coding agents and
low-code

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About this book

Every (v) mark in this book was checked on the date shown next to it; △ marks were not. Product names in this family change quickly — re-verify before relying on a number or a name.

Built from `course/` by `scripts/build_pdf.sh` (just `book`, or `nix build .#book` in the Nix sandbox); CI commits the PDF to `pdf/` on `main`. The source of truth is the repository: <https://github.com/h0ffmann/gcp-agentic-architect>.

Chapter 1

Orientation: how this exam is written and how to answer it

Exam objectives: all · section weight —. Quiz: just quiz --lesson 00. Case: just case 00.

1.1 On the exam

- Google Professional exams are **scenario elimination**: a company, a constraint, "what should you do?", one best answer. Two options are usually wrong products; one is a right product used wrongly; one is right. Find the constraint word first (cost, least ops, data residency, latency, audit, no code).
- **Newer names win**. If two options describe the same thing under old and new names (Agent Engine / Agent Runtime, Vertex AI Search / Agent Search, Vector Search 2.0 / Agent Retrieval), the exam wants the current name.
- **Managed beats custom** unless the scenario says otherwise (custom container, sidecar, GPU, on-prem). "Least operational effort" → Agent Runtime; "existing container pipeline" → Cloud Run; "node-level control" → GKE.
- **Policy beats prompt**. Any option that enforces security through the system prompt is wrong when a policy product exists (Agent Identity/PAB, Agent Gateway, Model Armor, HITL).
- **Protocol split**: MCP = agent ↔ tools/servers; A2A = agent ↔ agent.
- 3 hours, beta-length pool. Budget ~2 min per question, flag and move on, never leave blanks (no negative marking on Google exams — verify in the Pearson instructions on the day).

1.2 Read (fetched 2026-09-15)

- EXAM-BRIEF.md (this repo) — the guide as a checklist and the 28 tools.
- [Agent Platform overview](#) — the four pillars; every product on the exam is placed here.
- [How beta exams work](#).

1.3 Verified today

Product	What it does
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Exam	MCQ 3 h, Pearson, Sept 8-30; labs < 5 h on Google Skills after passing, late Oct-Dec
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Weights	13 / 17 / 33 / 22 / 15
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1.4 Case — cases/–

No case for the orientation; run just smoke instead.

1.5 marola port

Not applicable. The orientation for marola is marola/AGENTIC-ARCHITECT-MAPPING.md.

1.6 Quiz

just quiz --lesson 00 — questions tagged lesson: "00" in questions/bank.jsonl.

1.7 Flashcards

questions/cards/00.tsv (front TAB back). Import into Anki as basic cards.

Chapter 2

Model selection: LLM vs SLM, self-hosted vs SaaS, OSS vs proprietary

Exam objectives: 3.1 · section weight 33%. Quiz: just quiz --lesson 01. Case: just case 01.

2.1 On the exam

- The exam's model question is never "which model is smartest". It is a constraint puzzle over **cost, security/residency, latency, and agent architecture** (guide 3.1).
- Tells: "data must not leave the VPC / on-prem" → self-hosted open weights (Model Garden) on GKE; "narrow task, high volume, low latency" → SLM; "broad reasoning, no ML ops team" → Gemini SaaS via Vertex; "many small routing decisions inside a multi-agent system" → SLM for routers/classifiers, LLM only for the planner.
- Model Garden is the catalogue (200+ models, Google, partner, open); it is where the exam expects you to find open weights.
- Cost levers named in docs: model size, context length, caching, batch vs on-line. A right answer often pairs a small model with a bigger one, not one model for everything.
- Distractor pattern: "fine-tune the largest model" when the constraint was latency or cost.

2.2 Read (fetched 2026-09-15)

- [Agent Platform: agents overview](#) — Model Garden described as 200+ foundation models (v).
- [Exam guide 3.1](#) — the exact wording of the selection axes (v).

2.3 Verified today

Product	What it does
Model Garden	Library of 200+ foundation models from Google, partners and OSS for discovery
Gemini LLMs	Google's proprietary models via Vertex / Gemini API

2.4 Case — cases/01-model-selection

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

2.5 marola port

marola already runs local Ollama models by default and treats every cloud model as opt-in (azure/ module pattern). The port is a documented decision matrix in docs/AGENTIC-ARCHITECT-MAPPING.md §3.1 mapping marola's LlmClient implementations to the four axes; no code.

2.6 Quiz

`just quiz --lesson 01` — questions tagged lesson: "01" in questions/bank.jsonl.

2.7 Flashcards

questions/cards/01.tsv (front TAB back). Import into Anki as basic cards.

Chapter 3

ADK fundamentals: agents, tools, runners, local run

Exam objectives: 3.1 · section weight 33%. Quiz: just quiz --lesson 02. Case: just case 02.

3.1 On the exam

- ADK vocabulary the exam uses: **Agent** (instruction + model + tools), **tools** (Python functions, MCP tools, built-ins), **Runner** (drives a session, yields **events**), **session service** (in-memory locally, Vertex/Agent Runtime in production).
- Local loop: `adk web <agents_folder>` opens the dev UI on localhost:8000 with an **Eval tab**; `adk run` for the CLI. Nothing is billed until you deploy or call a hosted model.
- Deploy paths: `adk deploy docker`, `adk deploy cloud_run`, `adk deploy agent_engine` (Agent Runtime). The exam pairs each with a requirement (see lesson 08).
- ADK is **model-agnostic** and open source; the exam says "open-source libraries (e.g., ADK)". A distractor will propose a proprietary console product for code-first work.
- Multi-agent primitives (lesson 06): `SequentialAgent`, `ParallelAgent`, `LoopAgent`, LLM-driven transfer between sub-agents.

3.2 Read (fetched 2026-09-15)

- [adk-docs](#) · [adk-python README](#) — commands and deploy targets (v).
- [Memory Bank quickstart with ADK](#) — shows the local-then-Runtime path (v).

3.3 Verified today

Product	What it does
ADK adk deploy	Open-source, code-first, model-agnostic framework (Python, Java) for agents and docker --with_ui, cloud_run --with_ui, agent_engine

3.4 Case — cases/02-adk-fundamentals

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

3.5 marola port

marola's Kyo pipeline is already agent = instruction + tools + runner. The MIP adds a Session trait so events are recorded the ADK way (author, function_call, function_response), which lessons 03 and 07 need. Local impl: FileSessionStore under .tmp/.

3.6 Quiz

just quiz --lesson 02 — questions tagged lesson: "02" in questions/bank.jsonl.

3.7 Flashcards

questions/cards/02.tsv (front TAB back). Import into Anki as basic cards.

Chapter 4

Sessions, Memory Bank, managed sessions; Agents CLI skills

Exam objectives: 3.1 · section weight 33%. Quiz: just quiz --lesson 03. Case: just case 03.

4.1 On the exam

- **Session** = one conversation's ordered events, scoped to a user id. **Memory Bank** = long-term facts *generated* from sessions (GenerateMemories) and fetched in later sessions, scoped by user (and optionally agent). The exam tests that you know which is which.
- Sessions and Memory Bank are Agent Runtime features (**Agent Platform Sessions / Memory Bank**); you can create a Runtime instance as a session store **without deploying code**.
- ADK wiring: VertexAiSessionService + a memory service; a callback such as add_session_to_memory triggers extraction at session end.
- Distractors: Firestore/Memorystore hand-rolled stores when "managed" is required; "bigger context window" as a substitute for cross-session memory.
- **Agents CLI**: scaffolds a project (agents-cli-manifest.yaml), installs skills into ~/.agents/skills (Antigravity sees them), gives scaffold/eval/deploy skills and a Skill Registry skill; the guide names "plugins and agent vs human mode" — expect a question on human-in-the-loop vs autonomous execution of CLI workflows (△ exact semantics unverified).

4.2 Read (fetched 2026-09-15)

- [Memory Bank](#) — CreateSession → AppendEvent → ListEvents → GenerateMemories (v).
- [Sessions with ADK](#) — Runtime instance as session store, no deploy needed (v).
- [Agents CLI community tutorial](#) (v as a secondary source; △ official doc).

4.3 Verified today

Product	What it does
Agent Platform Sessions	Stores per-user conversation events; source of truth for context
Memory Bank	Generates and serves long-term memories from sessions; scope key
Agents CLI	scaffold/eval/deploy skills, manifest, Skill Registry access

4.4 Case — cases/03-sessions-memory

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

4.5 marola port

Session trait (lesson 02) + MemoryStore trait with a local FileMemoryStore that runs a deterministic fact extractor (reuse MIP-0039's fact guard vocabulary) at session close. Cloud impl in gcp/ would be Memory Bank; written-not-run.

4.6 Quiz

`just quiz --lesson 03` — questions tagged lesson: "03" in questions/bank.jsonl.

4.7 Flashcards

questions/cards/03.tsv (front TAB back). Import into Anki as basic cards.

Chapter 5

RAG: embeddings, Agent Retrieval / Vector Search, RAG Engine, reranking

Exam objectives: 3.2 · section weight 33%. Quiz: just quiz --lesson 04. Case: just case 04.

5.1 On the exam

- Names: **Agent Retrieval** = **formerly Vector Search 2.0** — collections of JSON objects, filter by vector *and* payload, built-in embedding population or bring-your-own embeddings, usage-based or resource-based pricing. **Vector Search 1.0** = the earlier ANN index (still listed). **RAG Engine** = managed end-to-end RAG pipeline. **Agent Search** (formerly Vertex AI Search) = enterprise search/grounding over connected data (Section 1).
- Pipeline questions test where quality is lost: chunking → embedding model choice → similarity scoring (cosine/dot) → top-k → **reranking** → prompt assembly. "Right docs retrieved but wrong ones used" → reranker. "Relevant docs never retrieved" → embeddings/chunking/hybrid search.
- Evaluation of retrieval is separate from evaluation of the answer (lesson 07).
- Cost tells: small workload → usage-based pricing; tuned performance → resource-based.

5.2 Read (fetched 2026-09-15)

- [Agent Retrieval \(formerly Vector Search 2.0\)](#) (v).
- [Google-Cloud-AI/agent-platform samples](#) — names RAG Engine, Agent Search, Vector Search 2.0 side by side (v).

5.3 Verified today

Product	What it does
Agent Retrieval	Unified vector + payload storage and filtering; collections; BYOE or built-in
Vector Search 1.0	Earlier ANN index product; Agent Retrieval keeps its performance
RAG Engine	Managed RAG over enterprise datasets
Agent Search	Semantic search engines over your data (ex Vertex AI Search)

5.4 Case — cases/04-rag-rerank

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

5.5 marola port

marola's knowledge/ store (MIP-0055) is the local Agent Retrieval; add a Reranker step to KnowledgeStore.query with a deterministic lexical reranker and record retrieval@k in the benchmark ledger. Cloud impl: Agent Retrieval collection as a KnowledgeStore.

5.6 Quiz

just quiz --lesson 04 — questions tagged lesson: "04" in questions/bank.jsonl.

5.7 Flashcards

questions/cards/04.tsv (front TAB back). Import into Anki as basic cards.

Chapter 6

MCP, Agent Registry, Google Cloud MCP Servers, Agent Identity for tools

Exam objectives: 3.2 · section weight 33%. Quiz: just quiz --lesson 05. Case: just case 05.

6.1 On the exam

- **MCP** connects an agent to tools/servers (JSON-RPC; tools/list, tools/call). **Google Cloud MCP Servers** are managed MCP endpoints for Google services; custom MCP servers wrap your APIs, managed databases, or third-party SaaS.
- **Agent Registry** is the organisation's catalog of agents, tools and MCP servers: automatic registration from Agent Runtime, manual registration for external hosts; Agent Gateway reads it to enforce access.
- **Agent Identity** gives each agent a unique attested identity (SPIFFE ID; mTLS + DPoP) instead of a shared service account; permissions are IAM roles on that principal; **legacy bucket roles cannot be granted**. Audit logs show agent and user.
- Question shapes: "agents share a service account, auditors unhappy" → Agent Identity; "third-party agent must be discoverable" → manual registration; "which tools may this agent call" → registry + gateway policy, never the prompt.

6.2 Read (fetched 2026-09-15)

- [Agent Identity overview](#) (v).
- [Agent Registry automatic registration](#) (v).
- [MCP spec](#).

6.3 Verified today

Product	What it does
Agent Identity	Per-agent managed identity, SPIFFE-based, bound tokens, auth m
Agent Registry	Central catalog; automatic from Runtime; manual for external; A
Google Cloud MCP Servers	Managed MCP servers for Google Cloud services

6.4 Case — cases/05-mcp-server

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

6.5 marola port

marola's CLI already exposes an MCP tool server. Port: a tools/registry.json allowlist read by the server (registry) and enforced by ai-jail deny rules (identity boundary). This is exactly the case's shape.

6.6 Quiz

just quiz --lesson 05 — questions tagged lesson: "05" in questions/bank.jsonl.

6.7 Flashcards

questions/cards/05.tsv (front TAB back). Import into Anki as basic cards.

Chapter 7

Multi-agent: sequential / parallel / graph, A2A, handoffs, Agent Runtime, policies

Exam objectives: 3.3 · section weight 33%. Quiz: just quiz --lesson 06. Case: just case 06.

7.1 On the exam

- Composition: independent subtasks → **parallel**; dependent steps → **sequential**; conditional branching and retries → **graph/loop**. Latency questions want parallel fan-out then sequential fan-in.
- **A2A** is for agent-to-agent delegation across teams/frameworks: agent cards for discovery, tasks for delegation. **MCP** is not the answer to "two agents from different vendors".
- Coordination products: Agent Identity (who), Agent Registry (what exists), Agent Runtime (where it runs), agent policies via Gateway (what it may do). A2A agents on Runtime register automatically; Runtime traffic goes through Gateway automatically.
- Cross-project tool calls from a Runtime agent: register the destination in the gateway project, Agent-to-Anywhere mode only.
- Distractor: Cloud Scheduler / Pub/Sub for request-time orchestration.

7.2 Read (fetched 2026-09-15)

- [Agent Registry automatic registration](#) — A2A on Runtime (v).
- [Agent Gateway setup](#) — cross-project destinations (v).
- a2a-protocol.org (v link).

7.3 Verified today

Product	What it does
A2A	Open standard: agents declare identity/capabilities (agent card) and ex
Agent Runtime	Managed hosting; routes through Gateway; egress + ingress modes
ADK workflow agents	SequentialAgent, ParallelAgent, LoopAgent

7.4 Case — cases/06-multi-agent-a2a

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

7.5 marola port

Recommender → Reviewer in marola is a two-agent sequential chain; the case's parallel fan-out maps to running the three ocean-data lookups concurrently in Kyo, and the A2A card is a `/.well-known/agent.json` on the MCP server. `cases/06` is the reference implementation.

7.6 Quiz

`just quiz --lesson 06` — questions tagged `lesson: "06"` in `questions/bank.jsonl`.

7.7 Flashcards

`questions/cards/06.tsv` (front TAB back). Import into Anki as basic cards.

Chapter 8

Evaluation: evalsets, golden data, Gen AI evaluation service, autoraters, CI

Exam objectives: 4.1 · section weight 22%. Quiz: just quiz --lesson 07. Case: just case 07.

8.1 On the exam

- **ADK evaluation:** `.evalset.json` of eval cases (user turns, expected tool trajectory, reference response); `adk eval <agent> <evalset>` with an eval config; criteria include tool-trajectory match, `response_match_score` (default threshold 0.8), rubric-based response/tool-use quality, and `safety_v1/trajectory-quality` that delegate to the **Agent Platform Eval SDK** and need a project. `adk conformance` = regression against a golden baseline.
- **Gen AI evaluation service** (Agent Platform, Optimize pillar) = managed model-based metrics at scale; **custom autoraters** when the rubric is bespoke.
- Test-set design: golden data + prompts + **edge cases**; evaluate **response and retrieval** separately; continuous evaluation in CI and on production samples (drift).
- Question tells: "tool order" → trajectory; "helpfulness at scale without writing a scorer" → evaluation service; "regression gate on each PR" → evalset in CI; "quality dropped after data refresh" → retrieval evaluation.

8.2 Read (fetched 2026-09-15)

- [ADK evaluate](#) · [criteria](#) (v).
- [adk_eval_starter](#) — runnable workshop (v).

8.3 Verified today

Product	What it does	Limits / n
ADK eval	evalset files, CLI + web Eval tab, criteria, conformance	response_
Gen AI evaluation service	Managed evaluation on Agent Platform	preview p
Custom autoraters	LLM-as-judge you write	—

8.4 Case — cases/07-evalset

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

8.5 marola port

MIP-0040's reviewer-judge is a custom autorater; add an evalset/ directory in marola's shape (user, expected_tools, reference) run by just benchmark, with trajectory exact-match and a fuzzy response threshold — the case is the runner.

8.6 Quiz

just quiz --lesson 07 — questions tagged lesson: "07" in questions/bank.jsonl.

8.7 Flashcards

questions/cards/07.tsv (front TAB back). Import into Anki as basic cards.

Chapter 9

Deployment: Agent Runtime vs Cloud Run vs GKE, scaling, cost

Exam objectives: 4.2 · section weight 22%. Quiz: just quiz --lesson 08. Case: just case 08.

9.1 On the exam

- **Agent Runtime:** fully managed, Sessions/Memory Bank/code-execution/Example Store built in, automatic Registry and Gateway integration, `adk deploy agent_engine`. Choose for least ops and managed state.
- **Cloud Run:** your container, HTTP, scale-to-zero, `adk deploy cloud_run`; state is your problem. Choose for existing container pipelines or custom runtimes.
- **GKE:** sidecars, GPU node pools, node-level network policy, also the coding-agent sandbox. Choose for control.
- Scaling/cost tells: bursty + idle → serverless; steady high throughput with GPUs → GKE; "no platform team" → Runtime.
- Console recon (D7): open Agent Platform → Runtime, Registry, Gateway, Security pages read-only; note the exact menu names in this file.

9.2 Read (fetched 2026-09-15)

- [Agent Runtime · Scale](#) (v).
- [adk-python README deploy section](#) (v).

9.3 Verified today

Product	What it does
Agent Runtime	Managed; sessions, memory, code execution, Example Store; observability bu
Cloud Run	Serverless containers
GKE	Kubernetes; sidecars, GPUs, network policy; sandboxes

9.4 Case — cases/08-runtime-choice

Console recon notes go here (D7):

- Agent Platform menu: ...
- Runtime page shows: ...
- Registry page shows: ...
- Gateway / Security page shows: ...

9.5 marola port

MIP-0008 Docker images = the Cloud Run path; add a `deploy/agent-runtime.yaml` manifest (written-not-run) next to the Dockerfile, and a decision note in the mapping doc.

9.6 Quiz

`just quiz --lesson 08` — questions tagged `lesson: "08"` in `questions/bank.jsonl`.

9.7 Flashcards

`questions/cards/08.tsv` (front TAB back). Import into Anki as basic cards.

Chapter 10

Troubleshooting and observability: loops, drift, tool latency, Cloud Trace / Logging

Exam objectives: 4.2 · section weight 22%. Quiz: just quiz --lesson 09. Case: just case 09.

10.1 On the exam

- Symptoms → tools: **reasoning loop** (same tool call repeated) → iteration caps + traces; **tool latency** → Cloud Trace spans per call; **drift** (quality slowly falls) → continuous evaluation on production samples vs golden set; **hallucination** → grounding/retrieval evaluation + Model Armor is *not* the fix; **system failure** → Cloud Logging with the ReasoningEngine resource filter.
- Gateway 403 "Egress request is not authorized" → agent identity roles + authorization policy; find failing calls in Logging.
- Cost optimisation: cache, smaller model for sub-steps, cap loops, shorten context, batch offline work.
- Observability names: Google Cloud Observability = Cloud Logging + Cloud Trace (+ Topology view on Agent Platform).

10.2 Read (fetched 2026-09-15)

- [Troubleshoot Agent Gateway](#) (v).
- [Agent Platform agents overview](#) — Optimize pillar names Cloud Observability and Topology (v).

10.3 Verified today

Product	What it does	Limits / no
Cloud Trace	Spans per LLM/tool call; latency attribution	—
Cloud Logging	Runtime logs under <code>aiplatform.googleapis.com/ReasoningEngine</code>	filter by lo

10.4 Case — cases/09-trace-triage

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

10.5 marola port

marola's Telemetry trait gets an OTLP-shaped span list; the case's loop and bottle-neck detectors become a just triage recipe over `.tmp/traces.jsonl`.

10.6 Quiz

`just quiz --lesson 09` — questions tagged lesson: "09" in `questions/bank.jsonl`.

10.7 Flashcards

`questions/cards/09.tsv` (front TAB back). Import into Anki as basic cards.

Chapter 11

Security and governance: OAuth 2.0 / Auth Manager, PAB, Agent Gateway, Model Armor, SDP, HITL

Exam objectives: 5.1 · 5.2 · section weight 15%. Quiz: just quiz --lesson 10. Case: just case 10.

11.1 On the exam

- **Agent Gateway** = policy enforcement point for all agent traffic; requires an authorization policy; IAP authenticates using IAM; **deploy IAP in dry-run first**; egress vs ingress modes; delegates content checks to **Model Armor**; can add natural-language context-aware controls.
- **Agent Identity + PAB**: per-agent principal; **principal access boundary** policies bound what the agent can ever reach; agent-to-tool **OAuth 2.0** flows are handled by the **auth manager**, which encrypts end-user credentials so the agent never sees them and the gateway decrypts at egress.
- **Model Armor**: prompt injection/jailbreak, responsible-AI filters, **Sensitive Data Protection**, image screening; **floor settings** (project-wide) vs **templates** (per request); inspect-only vs INSPECT_AND_BLOCK; 4 MB cap; document screening only in Gemini Enterprise; if unreachable the request proceeds unscreened; Gateway needs roles/modelarmor.calloutUser and modelarmor.user.
- **HITL** for consequential actions (money, deletion, external comms). **Identity propagation**: logs show agent and user.
- Distractor: "put the policy in the system prompt".

11.2 Read (fetched 2026-09-15)

- [Agent Gateway overview](#) · [set up](#) (v).
- [Agent Identity overview](#) (v).
- [Model Armor overview](#) · [Gateway integration](#) · [Agent Platform integration](#) (v).

11.3 Verified today

Product	What it does
Agent Gateway	Enforcement point; IAP/IAM authorization policies; Regi
Agent Identity / PAB / auth manager	Per-agent principal; boundaries; OAuth handled outside
Model Armor	Floor settings vs templates; inspect vs block; SDP inside
Sensitive Data Protection	DLP filters used by Model Armor

11.4 Case — cases/10-gateway-policy

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

11.5 marola port

ai-jail + .claude/settings.json deny rules + guard-*.sh hooks are marola's gateway; MIP-0039's deterministic fact guard is its Model Armor; the water-quality veto is HITL. The case's five denials are the acceptance test for docs/AGENTIC-ARCHITECT-MAPPING.md §5.

11.6 Quiz

just quiz --lesson 10 — questions tagged lesson: "10" in questions/bank.jsonl.

11.7 Flashcards

questions/cards/10.tsv (front TAB back). Import into Anki as basic cards.

Chapter 12

Coding agents: Antigravity, Claude Code on Google Cloud, MCP config, sandboxes, Agents CLI

Exam objectives: 2.1 · 2.2 · section weight 17%. Quiz: just quiz --lesson 11. Case: just case 11.

12.1 On the exam

- **Antigravity** (app, CLI agy, SDK): Gemini CLI's successor; keeps **Agent Skills, Hooks, Subagents, Extensions**; reads AGENTS.md / GEMINI.md as **rules**; skills are SKILL.md directories (.agents/skills project scope, global under ~/.gemini/...); terminal auto-execution = Request Review vs Always Proceed; non-workspace file access off by default.
- Customisation vocabulary for 2.2: **skill** (packaged know-how, auto-selected by description), **rule** (standing instruction), **hook** (runs at lifecycle events, e.g. before a tool executes), **plugin/extension** (bundles), **subagent** (delegated worker). Expect "which one is this?" questions.
- **Claude Code on Google Cloud**: same MCP/skills model; the exam cares that coding agents get tools via MCP servers and run in **secure sandboxes** (GKE, Cloud Workstations, Antigravity sandbox) when code is untrusted.
- Use cases named: refactor, optimise runtimes, patch application-layer vulnerabilities — always with tests + scanner + review.
- **Agents CLI** augments Antigravity to build/scale/govern/optimise deployed agents.

12.2 Read (fetched 2026-09-15)

- [Gemini CLI → Antigravity CLI transition \(Google Developers Blog\)](#) (v via tracked issue).
- [Codelab: Authoring Antigravity Skills](#) (v).

- antigravity.google/docs/cli — official; open on D9.

12.3 Verified today

Product	What it does
Antigravity	App + CLI (agy) + SDK; skills/hooks/subagents/extensions; rul
Claude Code on Google Cloud	Anthropic's coding agent used with Google Cloud (MCP, skills)
Cloud Workstations	Managed dev environments as sandboxes

12.4 Case — cases/11-hooks-and-skills

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

12.5 marola port

marola's .claude/ (hooks, rules, skills, settings) is the Section 2.2 material; MIP-0011/0013/0017 already document it. Port = a table in the mapping doc from Antigravity vocabulary to marola files.

12.6 Quiz

just quiz --lesson 11 — questions tagged lesson: "11" in questions/bank.jsonl.

12.7 Flashcards

questions/cards/11.tsv (front TAB back). Import into Anki as basic cards.

Chapter 13

Low-code: Gemini Enterprise, Agent Designer, CX Agent Studio, Agent Search, multimodal ingestion

Exam objectives: 1.1 · 1.2 · section weight 13%. Quiz: just quiz --lesson 12. Case: just case 12.

13.1 On the exam

- **CX Agent Studio:** state-based flows — **pages, transition routes** (intent/condition), **event handlers** (no-match, no-input, errors), webhooks. Deterministic multi-step collection → this.
- **Gemini Enterprise Agent Designer:** instruction-driven agents in the Gemini Enterprise workspace; **system instructions and prompt templates** (few-shot, chain-of-thought) in the console.
- **Connecting data:** Gemini Enterprise data connectors / **Agent Search** with source ACLs; **multimodal ingestion** (video, audio, images) must be processed into a searchable store before agents can use it.
- Honest gap: nothing here runs locally; learn the console vocabulary and the decision "state machine (CX Agent Studio) vs instruction agent (Agent Designer)".

13.2 Read (fetched 2026-09-15)

- [Exam guide 1.1-1.2](#) (v).
- [Agent Platform overview](#) — Agent Studio described as a low-code visual canvas (v; note the exam says *Agent Designer* and *CX Agent Studio*).

13.3 Verified today

Product	What it does
CX Agent Studio	Pages, routes, event handlers, webhooks
Gemini Enterprise Agent Designer	Instruction/prompt-template agents in Gemini Enterprise
Agent Search	Enterprise search/grounding with ACLs

13.4 Case — cases/12-lowcode-flow

Runs with no Google Cloud account and no dependencies beyond Python. Read the source; it is the concept reduced to 40 lines.

13.5 marola port

Honest gap in marola (no low-code surface). The case shows the state-machine shape only so the vocabulary sticks.

13.6 Quiz

just quiz --lesson 12 — questions tagged lesson: "12" in questions/bank.jsonl.

13.7 Flashcards

questions/cards/12.tsv (front TAB back). Import into Anki as basic cards.

Chapter 14

Mock exam 1

Timed, 60 questions, 120 minutes: `just quiz --build-mock 1` (fails loudly if any section lacks unused questions — write more first), then `just quiz --mock 1 --minutes 120`.

After scoring: `just quiz --stats` → predicted score and three weakest objectives → one `--weak` session per weak objective before the next lesson. Record the score in `STUDY-LOG.md`.

Weak-spot review comes before Section 1 on D12.

Chapter 15

Mock exam 2

Timed, 60 questions, 120 minutes: `just quiz --build-mock 2` (fails loudly if any section lacks unused questions — write more first), then `just quiz --mock 2 --minutes 120`.

After scoring: `just quiz --stats` → predicted score and three weakest objectives → one --weak session per weak objective before the next lesson. Record the score in `STUDY-LOG.md`.

Final review sheet: re-read every lesson's 'On the exam' block in one sitting, then the flashcards only.